

# Keep Eyes on the Sentence: An Interactive Sentence Simplification System for English Learners Based on Eye Tracking and Large Language Models

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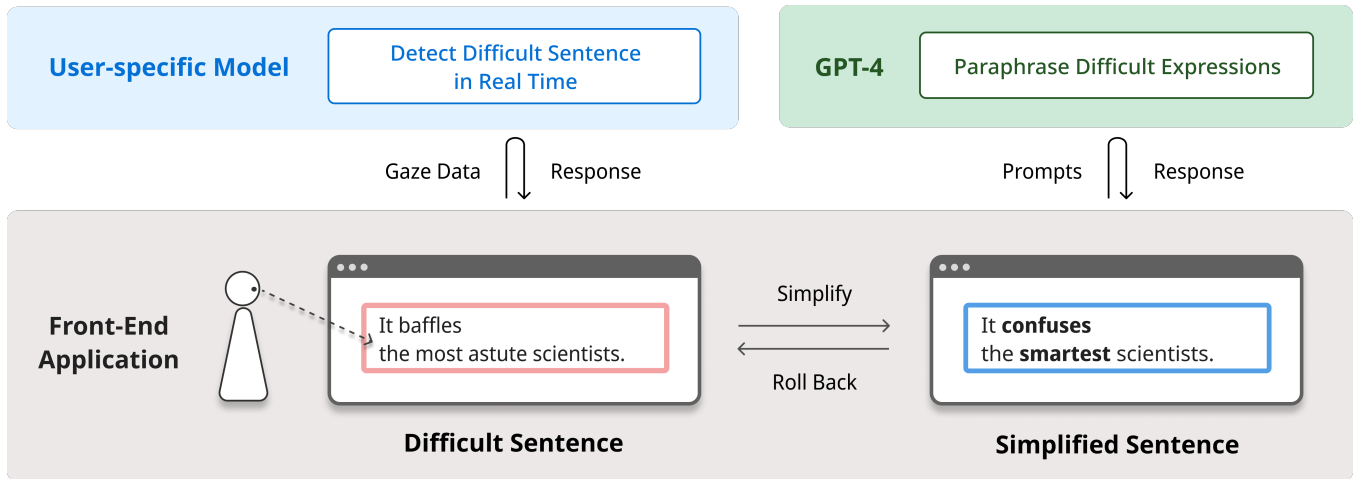


Figure 1: Overview of the proposed system.

## ABSTRACT

Language learners should read challenging texts regularly. However, using dictionaries or search engines to look up difficult expressions can be time-consuming and distracting. To address this, we have developed a system combining eye tracking with Large Language Models (LLMs) to simplify sentences automatically, allowing learners to focus on the content. The system incorporates user-tailored models that estimate users' comprehension of sentences using gaze data and sentence information. The system also features user-triggered simplification, resulting from iterative design improvements. We conducted a user study with 17 English learners where they read English text using either our system or a baseline

involving online dictionaries and search engines. Our system significantly improved both reading speed and comprehension, especially for complex sentences. The gaze-based simplification improved concentration on the content, allowing for an interruption-free reading experience. It could assist in daily reading practice, particularly for extensive reading focused on large volumes of text at a rapid pace.

## CCS CONCEPTS

• Human-centered computing → Interactive systems and tools; • Applied computing → Interactive learning environments.

## KEYWORDS

Eye tracking, human-computer interaction, sentence simplification, machine learning

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## 1 INTRODUCTION

Effective language learning involves regularly reading challenging materials, which can enhance reading fluency and vocabulary [26]. Learners often struggle with difficult expressions and have to refer to dictionaries when encountering difficult expressions, which can be time-consuming and distracting. Displaying simplified versions of sentences containing such expressions can enhance the learning experience [18]. This allows learners to check and acquire new expressions while keeping their focus on the content.

Large Language Models (LLMs) allow for sentence difficulty adjustment, simplifying English sentences while retaining original meanings, as confirmed by quantitative and human evaluations [10]. For personalized simplification, a mechanism to identify sentences needing simplification, based on individual comprehension levels, is crucial.

Eye gaze analysis offers a non-intrusive method to assess learners' comprehension during reading. Gaze patterns reveal the cognitive processes underlying reading comprehension. Previous research has employed machine learning to estimate word-level comprehension using eye gaze features [9, 15, 31]. However, focusing solely on single words may not fully capture understanding, as learners may struggle with multi-word expressions such as idioms and grammar. Therefore, it is necessary to estimate sentence-level comprehension that encompasses a range of linguistic components.

Based on these facts, we propose a gaze-driven sentence simplification system. Our system detects users' difficulties with a user-tailored machine-learning model that incorporates gaze features and linguistic features that reflect their reading behavior and sentence difficulty, respectively. When our system identifies users struggling with a sentence, it provides a simplified version, helping maintain content focus. Our system also includes iterative and user-initiated simplification, which were incorporated during the prototyping phase.

Finally, we conducted a user study with 17 English learners where they read English texts using either our system or a baseline approach involving online dictionaries and search engines. Participants using our system showed faster reading speeds and better focus. Compared to the existing approach, it also significantly enhanced comprehension and confidence in difficult sentences. Additional functions added through prototyping further aided in understanding challenging sentences.

## 2 RELATED WORKS

### 2.1 Text Readability and Simplification

In assessing text readability [8], various metrics like Flesch-Kincaid Grade Level (FKGL) [21] and Automatic Readability Index (ARI) [30] have been proposed, considering syllable count and character count, respectively. The Age-of-Acquisition (AoA) metric [22] also represents word-level difficulty. Pre-trained language models [17] also contribute to this evaluation. Gaze data has shown potential in readability assessment and enhancing NLP tasks [3, 12, 13].

Text simplification research utilizes neural network methods [20, 23, 24] and Large Language Models (LLMs) [10]. Utilizing these simplification technologies for learner-specific challenging sentences can optimize simplification [8].

### 2.2 English Learning and Eye Tracking

Eye tracking offers insights into reading behaviors [7, 28]. It has been integrated into interfaces for displaying additional information or word-for-word translations based on gaze movements [4, 5, 16, 19]. This technology also aids in estimating English proficiency [2] and assessing subtitle comprehension in English, providing translations for challenging words [11]. Despite their usefulness, word-for-word translations often neglect context. Research involving machine learning models to identify unfamiliar words via gaze movements has also been explored [9, 15, 31]. However, these models may miss complex idioms and grammar structures, focusing mainly on individual words. In a prior investigation, models were developed to identify and simplify complicated sentences [14], yet the study lacked a comprehensive system implementation and user study evaluations.

## 3 PROPOSED SYSTEM

This section details the implementation of our proposed system. Figure 2 illustrates the key components of the system. Our system requires a remote eye tracker with a laptop. With this setup, users read English text via a web application in the browser. For our study, we used Tobii Pro Nano eye tracker<sup>1</sup> mounted it to a MacBook Pro with a screen resolution of 1400 × 800 pixels.

### 3.1 Difficulty Detection

**3.1.1 Difficulty Detection Model.** We trained user-specific models to identify challenging sentences for individual users. We recruited 17 native Japanese-speaking university students and collected gaze data and the sentences they found difficult as ground truth for 90 minutes each. The dataset details are in Table 1. We used CatBoost [27] as a machine learning algorithm and trained a separate model for each user. Sequential Backward Selection (SBS) [1] was applied to refine features from the predefined set listed in Table 2. Our models' performance, with a 0.69 weighted F1-score, is shown in Table 1 and compares to previous similar models (0.556 [15], 0.75 [9]). These trained models are deployed on a difficulty detection server while the system is in use.

**3.1.2 Automatic Difficulty Detection.** Our system tracks eye movements in a web application at 60 Hz, recording gaze points with screen coordinates and timestamps. Using a stacked gaze data array, it detects fixations via Identification by Velocity-Threshold (IVT) [29], correlating these with corresponding words in texts using the browser's Document Object Model.

On fixation detection, a request with the user ID, sentence, and fixation data is sent to the difficulty detection server. This server processes data per user ID, extracting gaze and linguistic features, particularly those identified in the training phase. The user-specific model then evaluates sentence comprehension and informs the

<sup>1</sup><https://www.tobii.com/products/eye-trackers/screen-based/tobii-pro-nano>

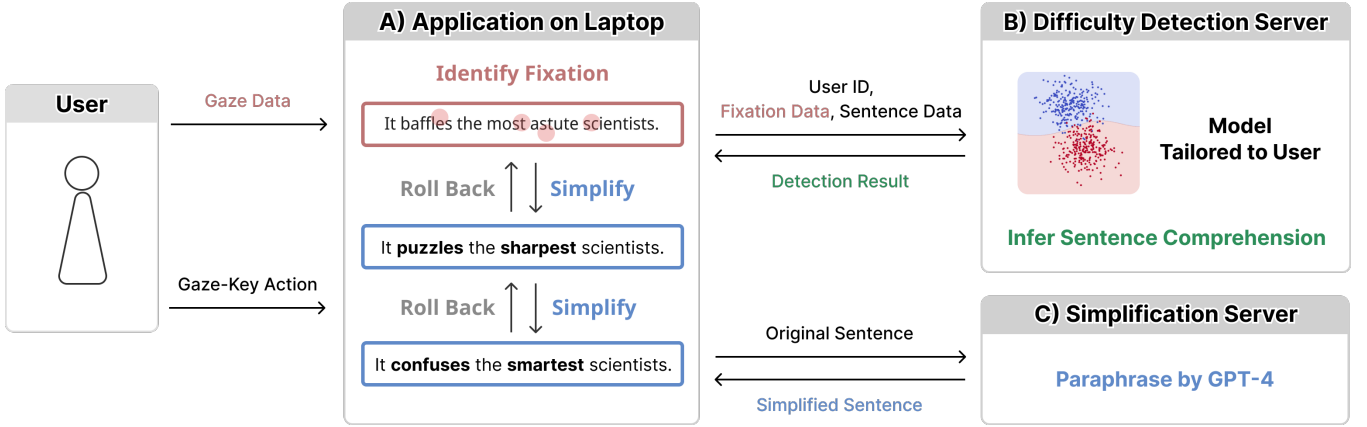


Figure 2: Our system comprises: A) an app that processes gaze data to identify fixations and simplifies or reverts sentences; B) a difficulty detection server deducing sentence comprehension from fixation and sentence data using a user-specific model; C) a simplification server receiving original sentences and returning simplified versions.

Table 1: Property of the dataset and the results of comprehension estimation.

|               | ID        | P01  | P02  | P03  | P04  | P05  | P06  | P07  | P08  | P09  | P10  | P11  | P12  | P13  | P14  | P15  | P16  | P17  | Avg. |
|---------------|-----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Dataset       | Sentences | 313  | 295  | 304  | 295  | 277  | 268  | 285  | 230  | 313  | 257  | 501  | 257  | 162  | 230  | 123  | 191  | 304  | 270  |
|               | Marked    | 133  | 64   | 69   | 127  | 62   | 93   | 101  | 50   | 48   | 49   | 127  | 97   | 59   | 102  | 47   | 52   | 86   | 73   |
| Weighted Avg. | Recall    | 0.64 | 0.71 | 0.86 | 0.58 | 0.78 | 0.67 | 0.72 | 0.84 | 0.89 | 0.81 | 0.74 | 0.69 | 0.67 | 0.67 | 0.69 | 0.73 | 0.75 | 0.73 |
|               | Precision | 0.64 | 0.66 | 0.85 | 0.58 | 0.70 | 0.65 | 0.72 | 0.82 | 0.88 | 0.85 | 0.79 | 0.69 | 0.65 | 0.68 | 0.70 | 0.68 | 0.76 | 0.69 |
|               | F1        | 0.64 | 0.68 | 0.86 | 0.58 | 0.69 | 0.65 | 0.72 | 0.83 | 0.87 | 0.82 | 0.81 | 0.69 | 0.66 | 0.67 | 0.69 | 0.69 | 0.69 | 0.69 |

Table 2: Features for estimating comprehension of sentences. No. 1 to No. 5 are gaze features, and No. 6 to No. 16 are linguistic features.

| No.   | Features                                             |
|-------|------------------------------------------------------|
| 1     | Maximum fixation duration.                           |
| 2     | Total durations of fixations.                        |
| 3     | Total fixation count.                                |
| 4     | Regressive fixation count.                           |
| 5     | Fixation duration per word (No. 3 divided by No. 6). |
| 6-8   | Total number of words, stopwords, and characters.    |
| 9     | Total number of content words.                       |
| 10    | Sentence length in pixels.                           |
| 11    | Total number of named entities.                      |
| 12    | Average named entities per word.                     |
| 13-14 | Sentence readabilities: ARI [30] and FKGL [21].      |
| 15-16 | Average word difficulty: AoA and word frequency [6]. |

client application about the user’s understanding, triggering text simplification if needed.

### 3.2 Simplification and Presentation

When our application detects difficulty, it sends a simplification prompt to GPT-4 through OpenAI’s API. The prompts used for simplification are detailed in Table 3, and they are based on previous

research [10]. Upon receiving the simplified sentence, the system replaces the original sentence with the simplified version.

### 3.3 Integrated Features via Prototyping

We prototyped a threshold-based system that simplifies sentences when a user takes more than a total of 500 milliseconds per word to read a sentence. We conducted testing of our prototype with four randomly selected participants, identified as P02, P06, P12, and P14.

**3.3.1 Gaze-Key Action and User-Triggered Simplification.** Participants suggested user-triggered simplification. Consequently, our system now allows users to initiate simplification by a *gaze-key action* (pressing Enter while looking at a sentence). The system calculates the targeted sentence using the average coordinates of recent user fixations.

Despite simplifications, participants found some sentences still challenging. Our system thus enables further simplification and reversion to original sentences via *gaze-key actions*. Repeated simplifications are possible with the Enter key, and rolling back to previous versions is achieved with the Space key.

**3.3.2 Highlighting Modified Words.** Participants expressed difficulty in grasping how a sentence is simplified, suggesting that highlighting modified words would be helpful. In response to this feedback, our system employs the Longest Common Subsequence algorithm [25] to identify the words that have been altered through simplification. Subsequently, the system simplifies sentences, highlighting the modified words in bold for clarity.

**Table 3: Prompt to simplify sentences.**

| Role   | Content                                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System | I want you to replace the user's complex sentence with simple sentence(s).<br>Keep the meaning the same, but make them simpler. Output only the simplified sentence(s). |
| User   | {The original sentence.}                                                                                                                                                |

**Table 4: Property of the texts used in the main study.**

| ID                  | T01 | T02 | T03 | T04 | T05 | T06 | T07 | T08 | T09 | T10 | Avg.  |
|---------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|
| Number of sentences | 6   | 7   | 8   | 7   | 7   | 6   | 6   | 8   | 8   | 8   | 6.9   |
| Number of words     | 129 | 134 | 159 | 136 | 81  | 98  | 102 | 136 | 173 | 179 | 132.7 |

## 4 MAIN STUDY

The goal of this study is to evaluate the effectiveness of our comprehensive system in aiding daily English reading. For the baseline, participants used online dictionaries and the Google search engine to look up the meanings of challenging expressions. This approach was chosen as it is widely adopted and offers diverse and reliable information. The study's participants were identical to those involved in the initial data collection phase. Each session of the study lasted 60 minutes, and participants received a compensation of \$15.

### 4.1 Tasks and Conditions

**4.1.1 Reading Session.** We used 10 English texts, each with about 130 words and seven sentences, extracted from a news website<sup>2</sup>. Participants read five texts through our system and the other five using the baseline approach. The aim was to understand the texts well enough to answer questions in the next session using either method.

**4.1.2 Question Answering Session.** Participants' comprehension was tested with three multiple-choice questions per text. They were not allowed to use our system or online dictionaries to newly look up words during this session but could review previously referred materials. In the baseline condition, they could revisit online dictionary pages, while with our system, they could review sentences simplified in the reading session.

### 4.2 Procedure

Participants were split into two groups: Group A and Group B. Group A initially used the baseline approach for five texts, then switched to our system for the next five. Group B did the opposite.

A 10-minute training session introduced each group to their respective systems. For our system, participants practiced on a sample text and answered related questions. For the baseline, they were instructed to use online dictionaries and Google search for difficult expressions, applying these tools to a sample text.

At the end of each QA session, participants marked difficult sentences, aiding in evaluating our system. After completing all texts, they responded to questions Q1–Q8 (Figure 3) on a seven-point Likert scale. These questions assessed our system against

the baseline (Q1–Q4) and its usability (Q5–Q8). We also collected Open-ended feedback.

### 4.3 Metrics

We measured *reading speed* as words per minute (WPM), starting the timer when participants clicked "start reading" and stopping it when they did "stop reading". We also analyzed *participants' comprehension of texts* through the accuracy of their responses to questions to assess the effectiveness of the system and baseline in understanding text content.

To evaluate the performance of difficulty detection, we report two metrics: *recall* and *precision*. We first categorized inference results for each sentence into four types: (1) *true positive* when difficulty is detected at least once for the sentence marked as difficult, (2) *false positive* when detected, but the sentence was not marked as difficult, (3) *false negative* when not detected, but it was marked as difficult, and (4) *true negative* when not detected, and it was not marked as difficult. Based on the categorized results, we calculate *recall* and *precision*.

## 5 RESULTS

### 5.1 Overall Performance

**5.1.1 Reading Speed.** The Wilcoxon signed-rank test revealed a significantly higher reading speed with our system compared to the baseline ( $p < .01$ ), as Table 5 indicates. Participants read about 20 more words per minute on average. However, P11 showed a slower speed due to challenges with complicated sentences, leading to confusion with our system. P11 mentioned in comment number 1, labeled as **C1**, "Facing tough sentences, I restart from the first sentence of the text for understanding. This process simplifies other sentences that I ended up reading."

**5.1.2 Participants' Comprehension of Text.** As Table 5 shows, there was no significant difference in question answering accuracy between our system and the baseline ( $p=.18$ ), with a 6 % higher accuracy for the baseline. This drop in accuracy with our system is due to difficulties in handling proper nouns and failure to simplify desired words, as highlighted in comments **C2** and **C3**: **C2**: "Understanding proper nouns, like buildings or people, was challenging with the proposed system, unlike the baseline which allowed for image searches", **C3**: "the system sometimes failed to simplify the words that I wanted to understand."

<sup>2</sup><https://www.newslevels.com>

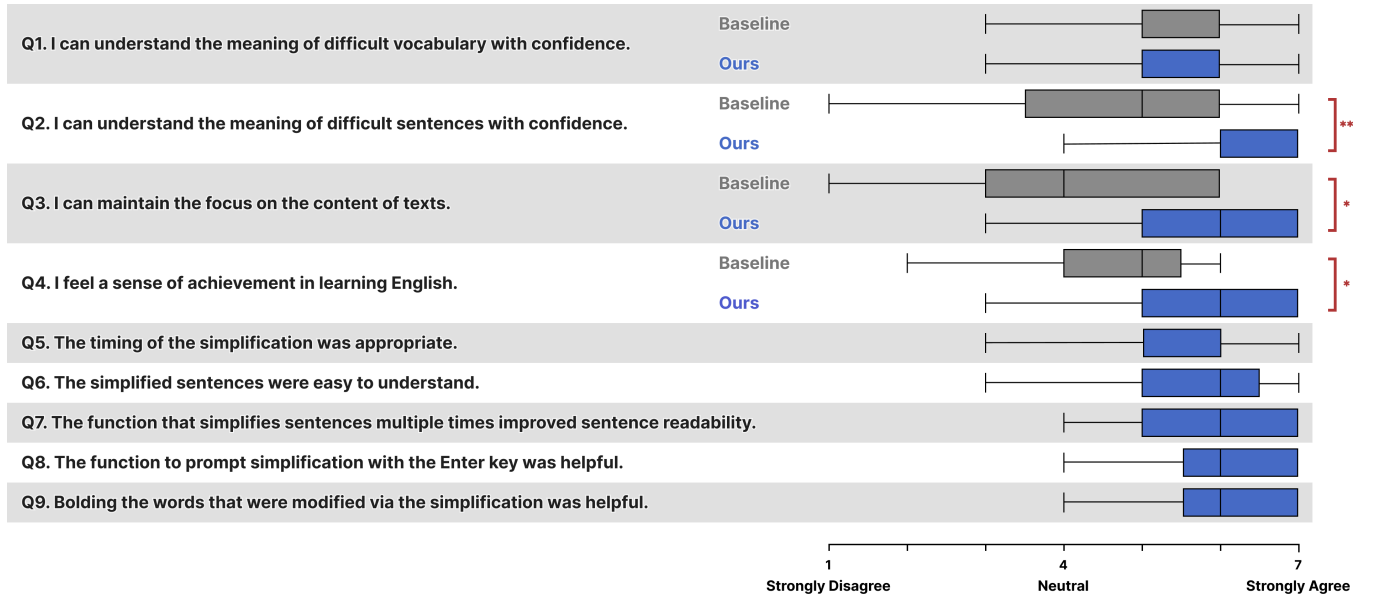


Figure 3: Main study Q&A with 17 participants, rated on a seven-point Likert scale. Asterisks in responses show significant differences using the Wilcoxon signed-rank test (\*:  $p < .05$ , \*\*:  $p < .01$ ).

**5.1.3 Subjective Ratings.** The subjective evaluation results in Figure 3 showed a preference for our system over the baseline for Q2, Q3, and Q4 ( $p < .05$ ), but no significant difference for Q1 ( $p = .60$ ), based on the Wilcoxon signed-rank test.

## 5.2 Performance of Difficulty Detection

Difficulty detection results in Table 5 show our models identified an average of 10 difficult sentences per group. The models had an average precision of 60% and recall of 59%. The drop in precision score from the training phase can be attributed to sentences being unintentionally detected through re-reading earlier sentences. Additionally, despite a decrease in recall, this does not indicate a decline in the model's performance. The reason is that the *gaze-key action* of the users caused all the false negatives before the model detected them.

## 5.3 Qualitative Feedback

**5.3.1 Positive Feedback.** Feedback indicated the system's functionalities were appreciated, especially the sentence-level simplification for understanding complex sentences. C4: "The system simplified all difficult expressions contained in a sentence altogether." (P05) Simplification was notably beneficial for long, structurally complex sentences. C5: "It split complicated sentences, preserving the original meaning." (P10) Participants reported an enhanced reading experience. C6: "The system improved focus over using online dictionaries, engaging me more effectively." (P03) Several participants noted the system's potential in English learning. C7: "It is more efficient for vocabulary building than dictionaries as it allows to learn new expressions in the context." (P02) C8: "It seems useful for extensive reading and understanding texts in the original language." (P01) C9: "It clarified figurative collocations that I misunderstood." (P14)

**5.3.2 Negative Feedback.** Participants were confused when simplification was triggered while intentionally taking time to re-read and understand the sentence. C10: "I had to read simplified sentences though I was still in the process of understanding the original one." (P09) C11: "I was upset as I felt I had to read before automatic simplification." (P06)

## 6 DISCUSSION

### 6.1 Comparison with Baseline Approach

The proposed system scored lower for understanding difficult vocabulary compared to the baseline (medians: 5 vs. 6, Q1 in Figure 3). However, it excels in understanding complex sentences, showing higher confidence scores than the baseline (medians: 6 vs. 5, Q2 in Figure 3). This system enhances reading focus and sense of achievement (Q3 and Q4 in Figure 3), and improves reading speed (Table 5).

In summary, our system is more effective for grasping texts' overall meaning. While dictionaries and search engines aid in vocabulary learning, they can be distracting. Our system aligns with extensive reading principles, as mentioned in C8, where learners are encouraged to read a substantial volume of materials, inferring difficult expressions' meanings from the context. Our system could support this approach by facilitating understanding and learning of complicated expressions without sacrificing reading speed.

### 6.2 Usability

Our system received favorable usability ratings, with medians of 6 for Q5-Q9. High ratings for Q7-Q9 highlighted that the functions added through the prototyping enhanced the reading experience. C12: "It changed the wording through second or third simplifications, which helped me confirm the meaning." (P17) C13: "Using the system, I could learn unfamiliar expressions in easier English, not in Japanese."

**Table 5: Demographic information of participants in our main study. Listed are the participants’ reading speeds in words per minute [WPM], the accuracy of the responses to the questions, and the performance of the difficulty detection. The mean and standard deviation (SD) are at the bottom. \*\*: significant difference when the Wilcoxon signed-rank test was applied ( $p < .01$ ).**

| Group | ID  | Gender | Age  | CEFR | Reading Speed [WPM] |        | QA Accuracy [%] |       | Actual Positive | Inference Performance [%] |           |
|-------|-----|--------|------|------|---------------------|--------|-----------------|-------|-----------------|---------------------------|-----------|
|       |     |        |      |      | Baseline            | Ours   | Baseline        | Ours  |                 | Recall                    | Precision |
| A     | P01 | Female | 23   | B2   | 42.3                | 57.6   | 60.0            | 73.3  | 11              | 45.5                      | 84.6      |
|       | P05 | Male   | 24   | B1   | 26.6                | 40.6   | 93.3            | 66.6  | 9               | 66.7                      | 52.9      |
|       | P07 | Male   | 24   | B1   | 73.6                | 93.3   | 73.3            | 73.3  | 17              | 53.0                      | 68.0      |
|       | P08 | Male   | 22   | B1   | 54.8                | 69.4   | 73.3            | 80.0  | 8               | 62.5                      | 61.5      |
|       | P09 | Male   | 25   | B2   | 39.1                | 60.3   | 80.0            | 100.0 | 7               | 43.0                      | 46.7      |
|       | P10 | Female | 31   | B1   | 88.8                | 142.4  | 100.0           | 100.0 | 10              | 60.0                      | 52.6      |
|       | P16 | Male   | 23   | B1   | 42.4                | 53.1   | 80.0            | 53.3  | 9               | 66.7                      | 52.9      |
|       | P17 | Female | 23   | B1   | 67.6                | 95.2   | 73.3            | 100.0 | 12              | 58.3                      | 57.4      |
| B     | P02 | Male   | 23   | B2   | 30.1                | 66.5   | 60.0            | 73.3  | 7               | 43.0                      | 50.0      |
|       | P03 | Male   | 24   | B2   | 66.4                | 92.9   | 86.6            | 73.3  | 8               | 62.5                      | 66.7      |
|       | P04 | Male   | 23   | B1   | 62.7                | 81.5   | 93.3            | 66.6  | 10              | 50.0                      | 90.9      |
|       | P06 | Male   | 25   | B1   | 40.2                | 47.4   | 80.0            | 73.3  | 10              | 40.0                      | 47.6      |
|       | P11 | Female | 22   | A2   | 86.4                | 50.1   | 93.3            | 73.3  | 6               | 83.3                      | 54.5      |
|       | P12 | Male   | 24   | B1   | 39.0                | 60.3   | 100.0           | 80.0  | 15              | 53.3                      | 60.0      |
|       | P13 | Male   | 22   | B2   | 49.4                | 44.3   | 100.0           | 86.6  | 11              | 81.8                      | 52.4      |
|       | P14 | Male   | 22   | B1   | 47.4                | 75.0   | 80.0            | 66.6  | 12              | 50.0                      | 60.0      |
|       | P15 | Male   | 22   | B1   | 18.6                | 31.2   | 80.0            | 80.0  | 10              | 70.0                      | 55.6      |
| Mean  |     |        | 23.7 |      | 51.5                | 69.3** | 84.6            | 78.4  | 10.1            | 59.0                      | 60.0      |
| ±SD   |     |        | ±2.1 |      | ±19.5               | ±26.1  | ±11.1           | ±12.6 | ±2.7            | ±13.1                     | ±11.8     |

(P10) Additionally, our gaze-based system facilitated a seamless reading experience without interruptions. **C14**: “*The gaze-based interaction allowed me to concentrate solely on the content since all I needed to do was read the texts.*” (P5)

There is room for improvements for simplification timing and readability. P11’s confusion with too early simplification led to a low Q5 score (**C11**). To mitigate this issue, the system could improve the performance of difficulty detection through online learning. P07’s feedback about simplification failing in paraphrasing target expressions resulted in a low Q6 score (**C3**). Our system should allow for specifying target expressions in prompts.

## 7 CONCLUSION

This paper presents a gaze-driven sentence simplification system designed to enable English learners to understand difficult sentences without distracting their attention. Our system includes user-specific models that detect difficulty from gaze patterns and sentence information. On the detection, it replaces the target sentence with an easier one generated by GPT-4. We conducted a user study where participants read English texts using our system and answered choice questions. The results demonstrated that the proposed system led to increased reading speed and improved content focus compared to the baseline approach, which involved the use of online dictionaries and search engines. While participants generally favored the baseline approach for understanding challenging vocabulary, particularly proper nouns, our system received higher

ratings for enhancing overall comprehension of difficult sentences. This suggests that our system is especially valuable in scenarios where overall sentence-level understanding takes precedence over in-depth word comprehension, such as extensive reading. For future work, we aim to incorporate a module to continuously improve difficulty detection and conduct a more comprehensive and extensive study of our system.

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